A Stacked Hybrid Ensemble of Swin-V2 and MViT-V2 for High-Performance Skin Disease Classification

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Abstract. Skin diseases pose a growing global health concern that is often underestimated and underprioritized in clinical practice. Timely and accurate diagnosis is vital for effective treatment. Recently, Deep Learning approaches, especially Convolutional Neural Networks (CNNs), have shown promise in automating the diagnostic process. However, as the number of disease classes increases, CNNs and traditional Visual Transformers (ViTs) generally struggle with visually similar diseases, resulting in decreased classification performance. To address these challenges, we propose a new approach that combines two advanced ViT architectures—Shifted Window Transformer (Swin) and Multi-Scale Vision Transformer (MViT)—and ensembles these models using a stacking technique to boost classification accuracy. Using a dataset of 19,559 images containing 23 skin disease classes from a Kaggle directory curated from Dermnet, our ensemble achieves 90% accuracy on unseen data, outperforming previous CNN-based approaches, which reached up to 81%. Additionally, the model attains an overall weighted F_1-score of 0.90 and recall of 0.90, demonstrating balanced performance for all classes. Importantly, our proposed ensemble model does not require computationally intensive devices to operate, making it accessible for low-resource clinical environments.

Keywords: Vision Transformers, Swin Transformer V2, Multiscale Vision Transformer V2, Ensemble Learning, Stacking Meta-Classifier, Skin Disease Classification, DermNet Dataset, Grad-CAM, Low-Resource Clinical Deployment

1 Introduction

Skin diseases pose a crucial global health challenge, with skin and subcutaneous conditions representing 4.02% of all years lived with disability (YLDs) in India alone [1]. Therefore, early diagnosis of skin conditions is crucial for achieving better outcomes, reducing the impact of the disease, and increasing survival rates [2].

Despite this, access to dermatological care remains limited. Studies show that only 24% of individuals seek professional help for skin conditions, while 90% of all cases are managed by general practitioners [3]. Furthermore, malignant conditions such as melanoma continue to increase, with over 10,000 deaths reported annually in the United States alone [4]. The World Health Organization has also recognized skin disease as the third most common cause of illness globally and a major cause of long-term disability [5].

Traditional practices depend significantly on manual assessments by dermatologists, which is time-consuming and vulnerable to human error [6]. This undermines the urgent need for accurate and efficient automated tools. With the rise of Artificial Intelligence (AI) in healthcare, computer vision has proven to be an excellent approach for medical image analysis and classification [7].

Convolutional Neural Networks (CNNs) have shown great success in processing image data and detecting localized patterns. However, they struggle to capture complex global context, which limits their classification accuracy to around 80% in multi-class tasks. To address these limitations, Vision Transformers (ViTs) have proven to be a powerful alternative. They utilize self-attention techniques that were originally popularized by natural language processing [8].

This paper presents an ensemble-based architecture that integrates two powerful ViT models—Swin Transformer and Multiscale Vision Transformer (MViT)—to enhance skin disease classification accuracy. The complementary characteristics of these architectures make them well-suited for ensemble learning.

The dataset used in this study includes 19,559 images across 23 disease classes, sourced from a Kaggle repository based on the DermNet database [9].

Finally, we evaluate model performance using standard classification metrics such as accuracy, precision, recall, F1-score, and class-wise breakdowns. We also utilize Grad-CAM visualizations to understand where the model's focus regions are, which provides insights into the decision-making process and enhances the system's transparency and trustworthiness.

2 Literature Review

The field of automated diagnosis for skin diseases has expanded rapidly, creating more efficient, accessible, and precise diagnostic tools. Early research in deep learning, especially Convolutional Neural Networks (CNNs), demonstrated significant improvements over traditional machine learning methods and can now match the diagnostic capabilities of human dermatologists [10]. Although foundational deep learning models like CNNs have achieved success, they often find it difficult to differentiate between visually similar skin conditions, resulting in performance ceilings around 80.95% [11].

To address this, the field has shifted towards Vision Transformers (ViTs), which are highly effective at capturing global context through self-attention [12]. The current state of this evolving field is moving towards synergistic models that merge the strengths of different architectures to achieve better performance. This started with ensembles of multiple CNNs that attained classification accuracy of 97.23% on the HAM10000 dataset [13].

Recently, hybrid models that combine CNNs and Transformer models have gained popularity due to their ability to utilize the local feature extraction of CNNs alongside the global context modelling of Transformers within a single, powerful framework [14]. This development has led to the most advanced methods: ensemble models composed of different Transformer architectures. The main idea is that even within Transformer families, different models acquire unique and complementary representations. One such model combined DINOv2, Swin Transformers, and ConvNeXt to achieve impressive accuracy on a 31-class skin disease classification task [15].

The idea and work presented in this paper advance the field. By creating a parallel ensemble of Swin-v2 and MViT-v2, our model combines two different yet complementary Transformer designs. Swin-v2 constructs its hierarchy through spatial partitioning with shifted windows [16, 17], while MViT-v2 employs semantic pooling to establish a multiscale feature pyramid [18]. This approach ensures efficient feature reuse and enhanced semantic understanding across multiple scales. This innovative architecture aims to generate a more comprehensive and powerful feature representation than either model alone, facilitating highly accurate and robust skin disease classification.

3 Methodology

This section explains the methodology for multi-class skin disease classification using Swin-V2 and MViT-V2 models, combined via a meta-learning-based stacking ensemble strategy. The methodology includes data preprocessing, data transformation, model architecture, training process, and evaluation metrics.

As shown in Figure 1, the pipeline initializes with data preprocessing and transformation. All input images are transformed into a standard resolution of 192×192 pixels, which is required by both the Swin-V2 and MViT-V2 architectures. Image quality is enhanced through contrast adjustment and sharpening for better visual clarity, enabling more effective feature extraction during model training. Since the dataset classes are heavily imbalanced, data transformation is applied to underrepresented classes by generating artificial variations. This not only manages the issue of class imbalance but also increases the dataset size, providing the models with increased data to learn from. The transformed dataset is then split into training, test, and validation sets in the ratio 70:20:10.

The Swin-V2 and MViT-V2 serve as the core architectures for feature extraction. These models are trained on a preprocessed and augmented image dataset, and their performance is validated using a separate validation set to ensure generalization and to test their ability to perform well on unseen data.

After training, the models are then evaluated using comprehensive performance metrics like accuracy, precision, recall (sensitivity), F1-Score, and the confusion matrix. These metrics provide a complete understanding of the model's performance across different classes and help assess the comprehensive reliability of the proposed framework.

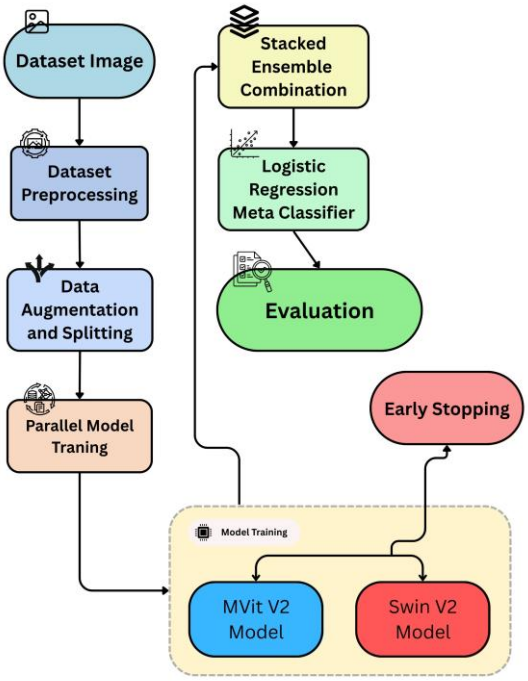

Fig. 1. Pipeline overview from data collection to model training.

3.1 Dataset

The dataset used in this study is acquired from the DermNet database, a reputable and clearly defined resource for dermatological image classification. DermNet offers clinical images categorized into 23 disease classes, examples of which are shown in Figure 2, making it a valuable standard for developing and evaluating deep learning models for medical image analysis.

For this research, a total of 19,559 images divided into 23 distinct categories were sourced from a Kaggle repository based on DermNet. Each category represents a type of dermatological condition. The distribution of images across these categories is shown in Figure 3, offering insight into the class imbalance within the dataset.

Fig. 2. Six example images from different disease classes in our 23-class DermNet dataset

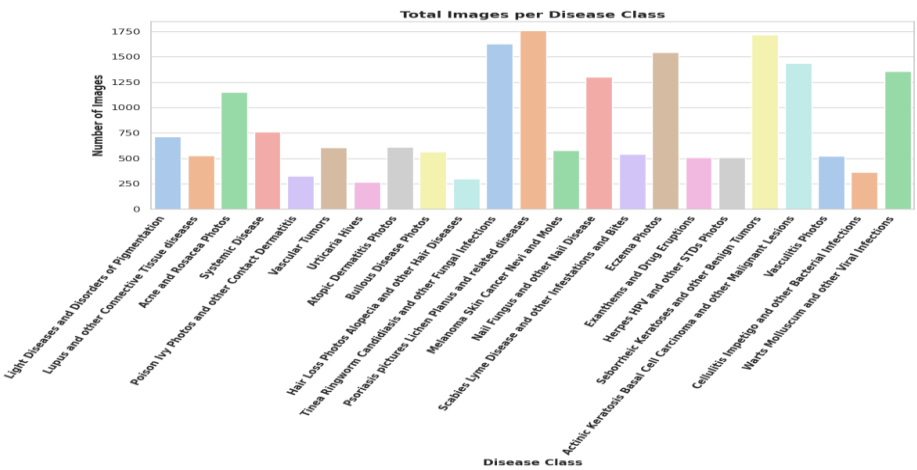

Fig. 3. Number of images per class.

3.2 Data Preprocessing and Transformation

The raw dataset comprises various dimensions and aspect ratios that must be standardised before training the models. Since the Swin-V2 and MViT-V2 architectures require inputs of 192x192x3, each image is converted accordingly to meet these specifications. We cannot resize the images directly without ensuring that important regions are preserved, as this could cause distortion and loss of critical visual features. To address this, centre cropping is applied first to retain the most relevant parts of the image, followed by resizing to the specified dimensions. This approach helps preserve essential information while maintaining model compatibility.

As shown in the table, the dataset is highly imbalanced, with some classes having significantly fewer images than others. Therefore, a well-structured data transformation strategy is employed to manage class imbalance and enhance model generalization. For classes with fewer than 1,000 images, samples are augmented to reach 1,500 images, while classes with more than 1,000 images are expanded to 2,000 images each.

The Data Transformation pipeline applies random rotations of up to ±15°, random horizontal flips, and adjusts brightness and contrast to simulate real-world variations in orientation, lighting, and color.

These transformations not only help balance the dataset but also introduce variety that prepares the model for real-world scenarios, such as variations in imaging conditions, lighting, and skin tones. This leads to a more balanced and effective classification framework that reduces bias and improves performance on unseen data.

3.3 Model Architecture

In this section, we will discuss the Transformer models used for training on the 23-class skin disease dataset. Since their introduction, Transformers have completely transformed vision tasks, especially with the rise of Vision Transformer (ViT), which has been widely used due to its ability to capture global context within images.

Swin Transformer: The Swin transformer provides an effective, hierarchical approach to visual understanding, making it suitable for tasks like skin disease classification. Unlike ViT, which processes fixed-size global patches, Swin Transformer starts with smaller patches and builds representations hierarchically. Instead of performing global self-attention, Swin applies self-attention within local windows, reducing computational cost. These windows are then shifted at each Transformer block layer to facilitate cross-window information exchange. This enables the model to capture global context.

Multiscale Vision Transformer (MViT): The MViT improves on the standard Vision Transformer's flat, single-scale design by introducing hierarchical, pyramidal structure. In this architecture, feature maps are gradually down sampled while increasing the number of channels at every stage. This allows MViT to process visual information at multiple resolutions. The self-attention technique is applied within increasingly coarser spatial resolutions. This lets the model efficiently capture both fine-grained local and global details and patterns [19]. MViT pools the keys and values before applying self-attention to optimize the performance even further. This reduces the higher computational cost of self-attention, allowing us to scale it to deeper layers and input higher resolution data without exceeding memory limits of our machine. This multiscale approach balances between computational efficiency and performance of the model. As a result, MViT offers a practical and scalable solution for complex visual understanding problems [20].

3.4 Model Ensembling – Stacking Meta Classifier using Logistic Regression

To further improve the performance, we used model ensembling using a stacking approach. In this method, multiple unique models are first trained on the dataset. Their individual outputs are then given to a meta classifier, which learns how to combine their predictions for the best results. Here we use logistic regression as the meta classifier. By assigning weights to the predictions of the base models, it then favors the models that perform well for specific types of inputs. This technique uses the strengths of different architectures, removing model biases and improving generalization [8]. This ensemble generates better robustness and accuracy compared to any individual model, especially when handling multi-class classification tasks such as skin disease diagnosis.

3.5 Evaluation Metrics

To evaluate model performance and robustness, we employ the following metrics:

Accuracy: Overall correctness on training and validation sets:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Precision and Recall (for class k): Precision measures the correctness of positive predictions:

$$\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}$$

and recall (sensitivity) measures detection of actual positives:

$$\mathrm{Recall_k} = \frac{\mathrm{TP_k}}{\mathrm{TP_k} + \mathrm{FN_k}}$$

F1-Score (for class k): The harmonic mean of precision and recall:

$$\mathrm{F1_k} = 2 \times \frac{\mathrm{Precision_k} \times \mathrm{Recall_k}}{\mathrm{Precision_k} + \mathrm{Recall_k}}$$

ROC AUC: Area under the Receiver Operating Characteristic curve, summarizing the trade-off between true positive rate (TPR) and false positive rate (FPR).

Precision–Recall Curve & Average Precision: Plot of precision vs. recall over varying thresholds; average precision is the area under this curve.

Confusion Matrix: Visual depiction of true vs. predicted labels across all classes.

Macro and Weighted Averages: For any per-class metric M_k

$$\text{Macro-}M = \frac{1}{C}\sum_{k=1}^{C} M_k$$

$$\text{Weighted-}M = \frac{1}{N}\sum_{k=1}^{C} n_k \; M_k$$

where C is the number of classes, n_k is the number of true samples in class k, and

$$N = \sum_{k=1} n_k$$

These metrics together provide a complete assessment of our ensemble's classification performance and robustness.

4 Results and Discussion

Both Swin-V2 and MViT-V2 performed exceptionally well during training. As shown in Figure 4 training accuracy of both models gradually increase with each epoch, ultimately reaching 98-99% by the end of 30th epoch.

Figure 4 shows the validation accuracy of both the models with respect to epochs; we can see that the Swin model achieves 87% validation accuracy and the MViT model achieves 86% validation accuracy which shows the model capability to learn patterns and predict the disease accurately.

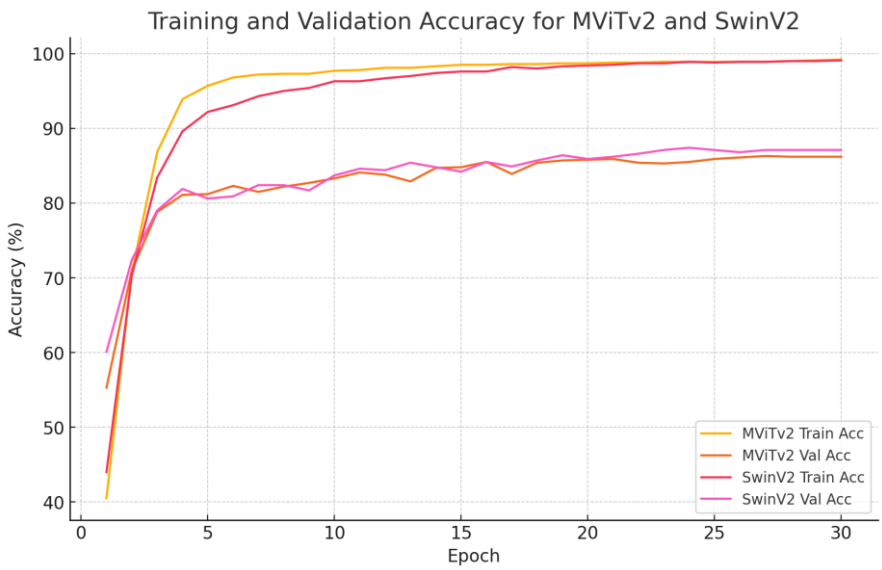

Fig. 4. Training accuracy and Validation accuracy vs. epoch for Swin and MViT models.

In Figure 5 we can see the training loss and validation loss with respect to epoch for both models. The training loss has drastically decreased from 2 to 0.01 and the validation loss has decreased from 1.5 to 0.75, which indicates steady convergence without severe overfitting.

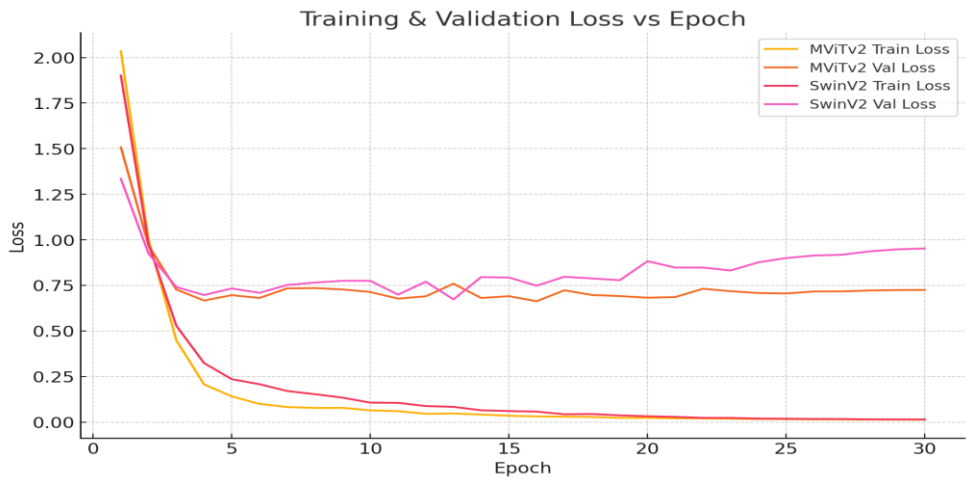

Fig. 5. Training and validation loss vs. epoch for both models.

Further in Figure 6 we can see the confusion matrix on the 10% unseen validation data to check the robustness of the overall model. The ensemble model was used to make predictions and generate the confusion matrix.

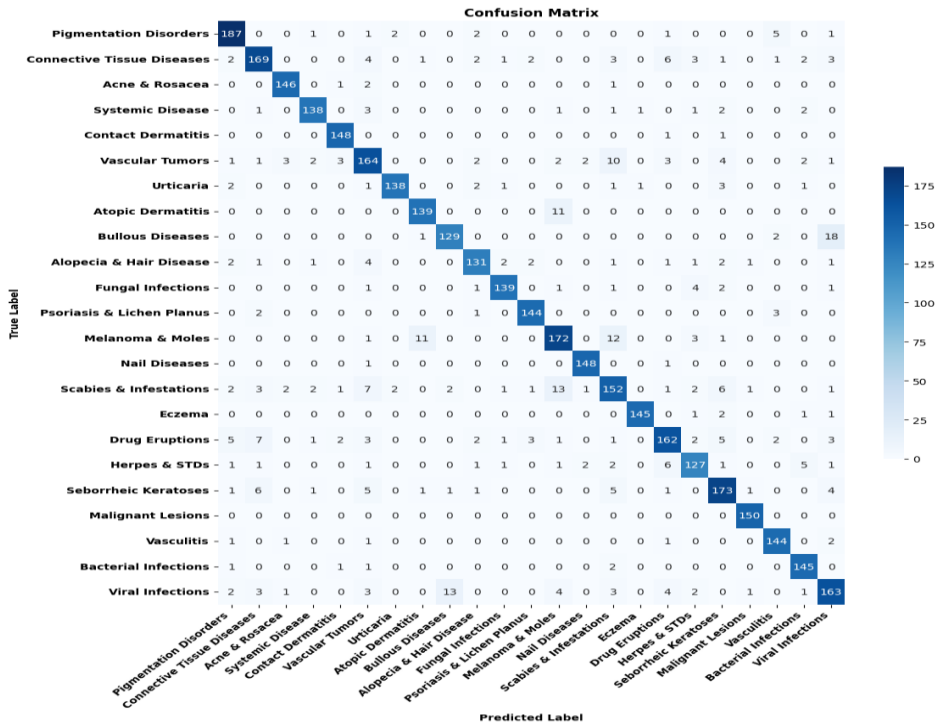

Fig. 6. Confusion matrix on the unseen 10% validation set using the ensemble model.

Table 1 summarizes the validation accuracy of our proposed model with several state-of-the-art approaches for direct comparison.

Table 1. Comparison of Top-1 Accuracy

Model	Validation Accuracy (%)
DenseNet201 [21]	68.97%
Ensemble of VGG16 and EfficientNet-B3 [22]	69.00%
Custom Convolution Transformer [23]	73.8%
Our proposed model: MViTv2 + SwinV2 Ensemble w/ Meta-Classifier	**90.0%**

Table 2 shows class-wise Precision, Recall, F1-score and support. The overall accuracy of the ensemble model reached 90% and the macro average and weighted average are identical (0.90), showing balanced performance across classes.

Table 2. Class-wise Precision, Recall, F1-score and Support

Class	Prec.	Rec.	F1	Supp.
Acne & Rosacea	0.90	0.94	0.92	200
Actinic Ker. & BCC	0.87	0.84	0.86	200
Atopic Dermatitis	0.95	0.97	0.86	150
Bullous Disease	0.95	0.92	0.93	150
Cellulitis & Impetigo	0.95	0.99	0.97	150
Eczema	0.81	0.82	0.81	200
Drug Eruptions	0.97	0.92	0.95	150
Hair Diseases	0.91	0.93	0.92	150
STDs	0.89	0.86	0.87	150
Pigmentation Disorders	0.90	0.87	0.89	150
Lupus & CT Diseases	0.95	0.93	0.94	150
Melanoma & Nevi	0.95	0.96	0.95	150
Nail Disease	0.83	0.86	0.85	200
Contact Dermatitis	0.97	0.99	0.98	150
Psoriasis & Lichen Planus	0.78	0.76	0.77	200
Scabies & Bites	0.99	0.97	0.98	150
Seborrheic Keratoses	0.86	0.81	0.84	200
Systemic Diseases	0.87	0.85	0.86	150
Fungal Infections	0.85	0.86	0.86	200
Urticaria	0.97	1.00	0.99	150
Vascular Tumors	0.92	0.96	0.94	150
Vasculitis	0.91	0.97	0.94	150
Viral Infections	0.81	0.81	0.81	200
Macro avg	0.90	0.90	0.90	3850
Weighted avg	0.90	0.90	0.90	3850

Outstanding performance was observed for Scabies & Bites (F1 = 0.98), Poison Ivy (F1 = 0.97), and Melanoma & Nevi (F1 = 0.95). Lower performance was observed for Eczema (F1 = 0.81), Psoriasis & Lichen Planus (F1 = 0.77), and Viral Infections (F1 = 0.81), indicating room for improved transformation and preprocessing.

Figure 7 shows Grad-CAM heatmaps on original images, highlighting regions that the model focuses on for classification. This provides deeper insight into the model's decision-making process and increases transparency.

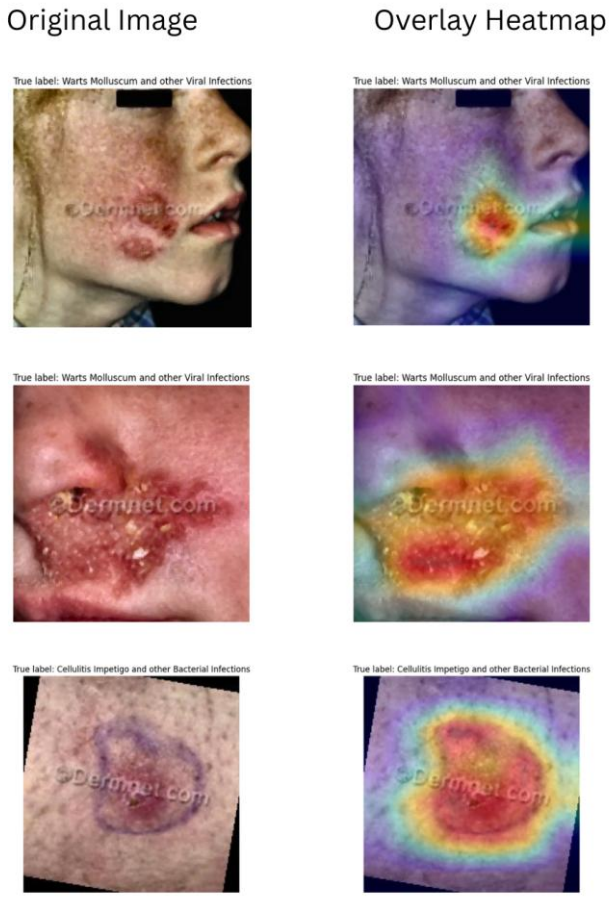

Fig. 7. Grad-CAM heatmaps showing model focus regions (original image left, heatmap overlay right).

5 Conclusion

The evolution and growth of Vision Transformers have made the automation of skin disease classification significantly easier, faster, and more accurate over the past couple of years. By leveraging the unique capabilities of multiple models like CNNs and Transformers or combining multiple Transformer models, we can achieve better results than either could on their own. Our approach is also suitable for low to medium end devices, making it much more accessible compared to heavier models that may offer slightly better performance but are far more computationally expensive.

However, despite the significant growth of computer vision, it still does not match the performance and reliability of a professional dermatologist. This gap can be set down to several factors, primarily the limited availability of high-quality data. Even when data is available, the presence of many visually similar diseases can confuse the model and lead to misclassification. As a result, models tend to perform better on datasets with fewer classes, where there is a lower chance of confusing two or more visually similar conditions.

Further betterments in this field will not only depend on architectural innovation but also resolving data related challenges. Increasing access to diverse, high resolution and verified datasets will be essential to further reduce the gap between AI based classification and professional diagnosis.

Ultimately, our work contributes to this ever-evolving field by proposing an ensemble of Swin-V2 and MViT-V2 models that offer a robust and efficient framework for accurate multi-class skin disease classification.

6 Availability Of Code

- https://www.kaggle.com/code/dhairyadoshi45/final-codes

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